BEYOND ACCURACY: WHAT HAPPENS WHEN AI MEETS REAL-WORLD DATA?

Who Am I?

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Amazon Marketplace Consultant – Denman International Ltd., UK

Educational Background

MSc in AI – Ulster University, UK

BSc in Computer Applications [Triple Main] - Mathematics, Statistics and Computer Science – Sacred Heart College, Thevara

My work spans AI, analytics and software development

- **Applied AI & Research**
  Wheat Rust AI — image classification, cross-dataset evaluation and explainability.
- **Business Intelligence & Analytics**
  Amazon PPC Intelligence Hub and ***Superstore Business Intelligence*** — advertising analysis, data pipelines and reporting.
- **Software & Community Platforms**
  Bloodline RP — developed and launched a live FiveM roleplay server with custom gameplay systems and an active player community.

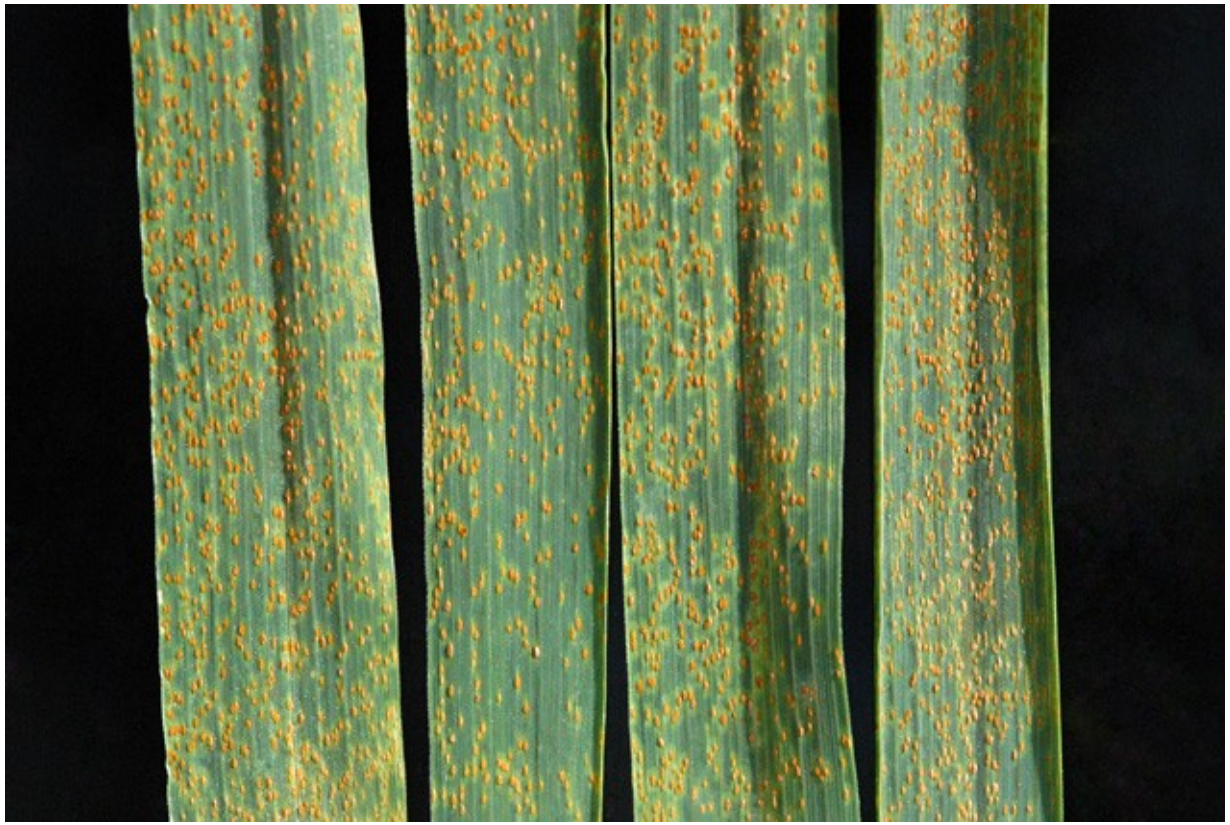
Where the story began: Wheat Rust?

- **What is Wheat Rust?**

-A fungal disease affecting wheat!

- **Could AI help recognise signs of disease from a leaf image**?

Wheat Rust?



Diseased Leaf



Healthy Leaf

One Crop! 3 Types of Rust

- **Leaf Rust**

  Also called ***Brown Rust***

- **Stripe Rust**

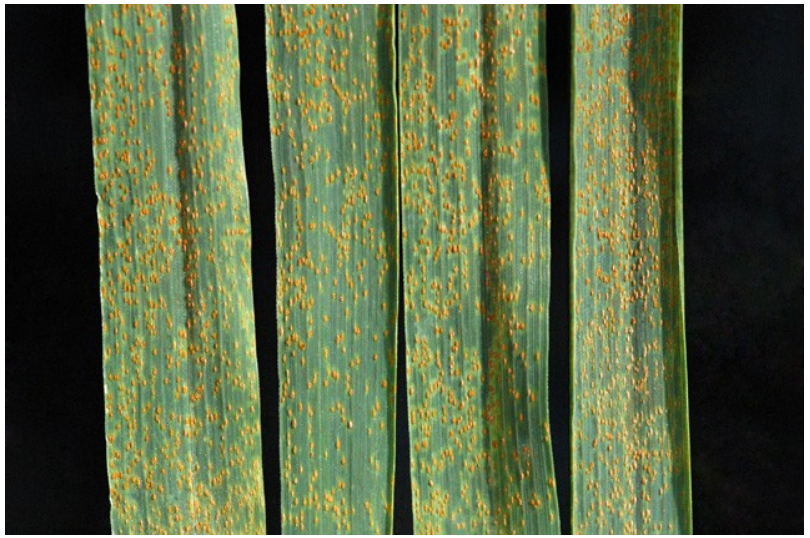
  Also called ***Yellow Rust***

- **Stem Rust**

  Also called ***Black Rust***

My classifier focuses on healthy leaves, brown rust and yellow rust.

One Crop! 3 Types of Rust



Brown Rust

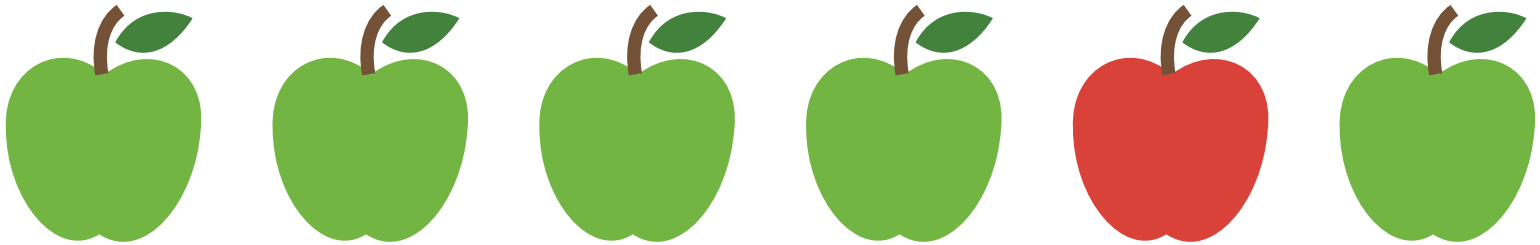


Black / Stem Rust



Yellow Rust

Can you spot the Odd one out?



Anomaly detection means identifying something that differ from pattern we expect

Why did I Show this?

My Wheat Rust Project

- The model learned from **healthy leaf images only**.
- Instead of asking "Is it Yellow/Brown Rust" for each new image, it asked: **“Does this leaf look different?”**
- But lighting, backgrounds and cameras can also cause differences.

Looking Different ≠ Diseased

My Msc Starting Point: Exploring Anomaly Detection

- During Masters Project, I explored:
  - Isolation Forest
  - Autoencoders
- My Question:

Could unusual image patterns help identify wheat rust?

- The Challenge

Would the model detect disease—or simply differences in appearance?

Why I revisited the evaluation

How the Evaluation Metrics have been calculated?

- Some original metric outputs used example labels rather than actual model predictions.
- I needed a clearer connection between each image, its true label and the model's prediction.
- **This became a priority in my later independent research**

Why I revisited the project

Making the connection between actual image and predicted image become priority:

- Which Image was evaluated?
- What was its correct label?
- What prediction the model produced?

Three questions guided my investigation

- **Evaluation:** Were predictions compared with the correct labels?
- **Generalisation:** Would the model work on a different dataset?
- **Explanations:** Which image regions influenced its predictions?

Teaching the model with labelled examples

Each training image had a label:

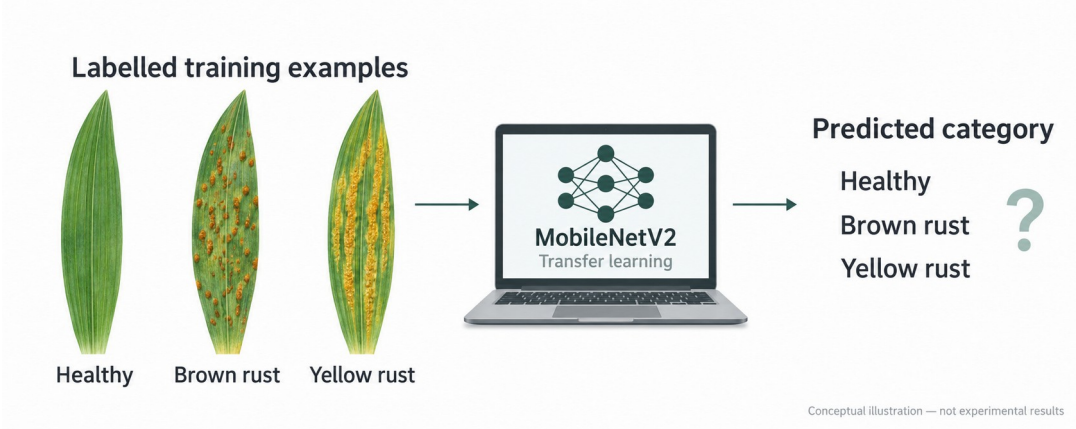
Healthy · Brown rust · Yellow rust

I used MobileNetV2 with transfer learning to build a classifier for these three categories.

Instead of asking "Is this unusual?",

the classifier asks

"Which category does this image belong to?"



What is transfer learning?

Start with a pretrained model, then adapt it to a new task.

- Pretrained visual features
- ↓
- Training on labelled wheat-leaf images
- ↓
- Three-class prediction

Question: How well the adapted model performs on new images?

Putting the classifier to the test

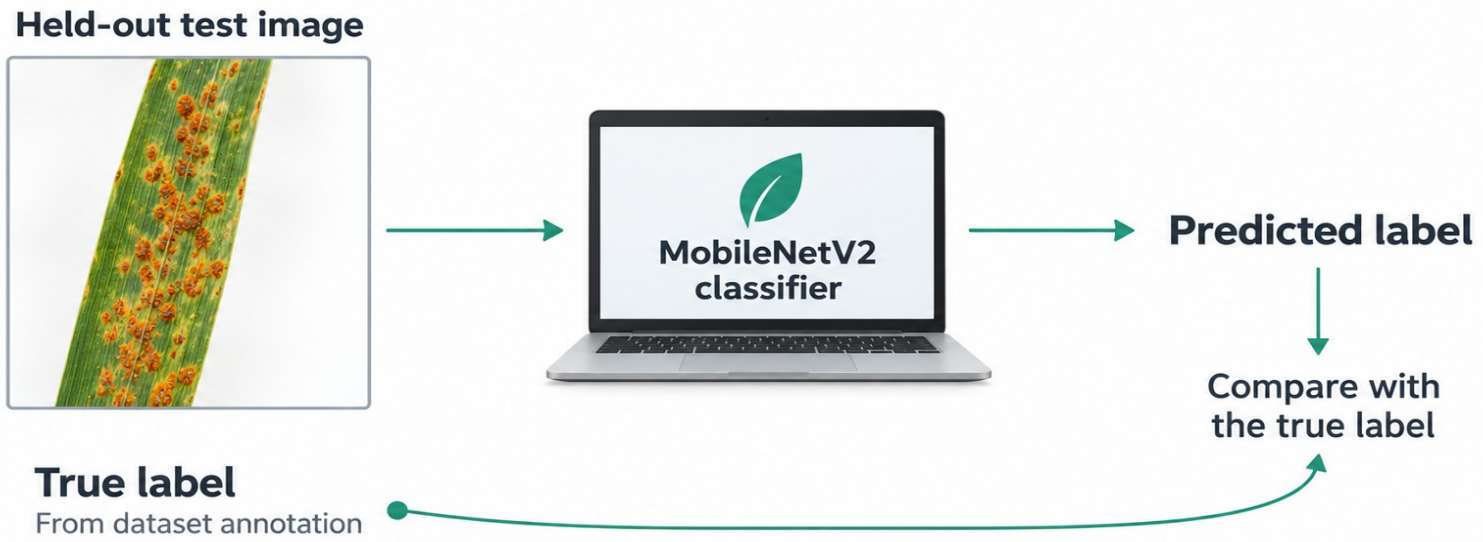


Internal validation: **737 images**

External evaluation: **489 images**

Which classes did it confuse?

Putting the classifier to the test



A promising result—but on which data?

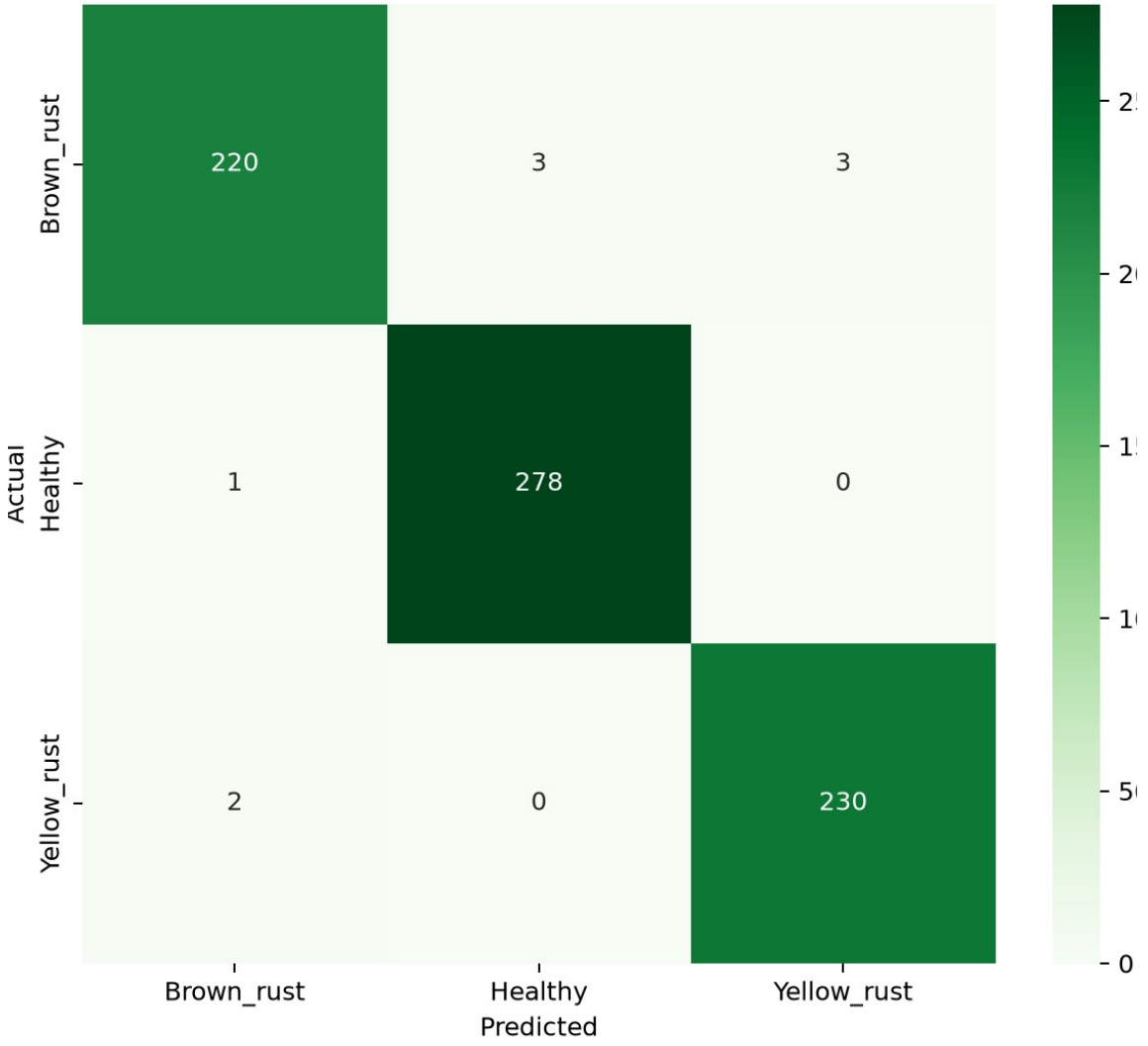
- 98.78%

Internal validation accuracy

- 728 / 737 images classified correctly

9 misclassifications

- Would this performance hold on images from another source?



Which classes did it confuse?

Actual ↓ / Predicted →	Brown rust	Healthy	Yellow rust
Brown rust	220	3	3
Healthy	1	278	0
Yellow rust	2	0	230

Same disease label. Different photographs.

Both labelled **brown rust**. Would the model recognize both



Org Validation Dataset

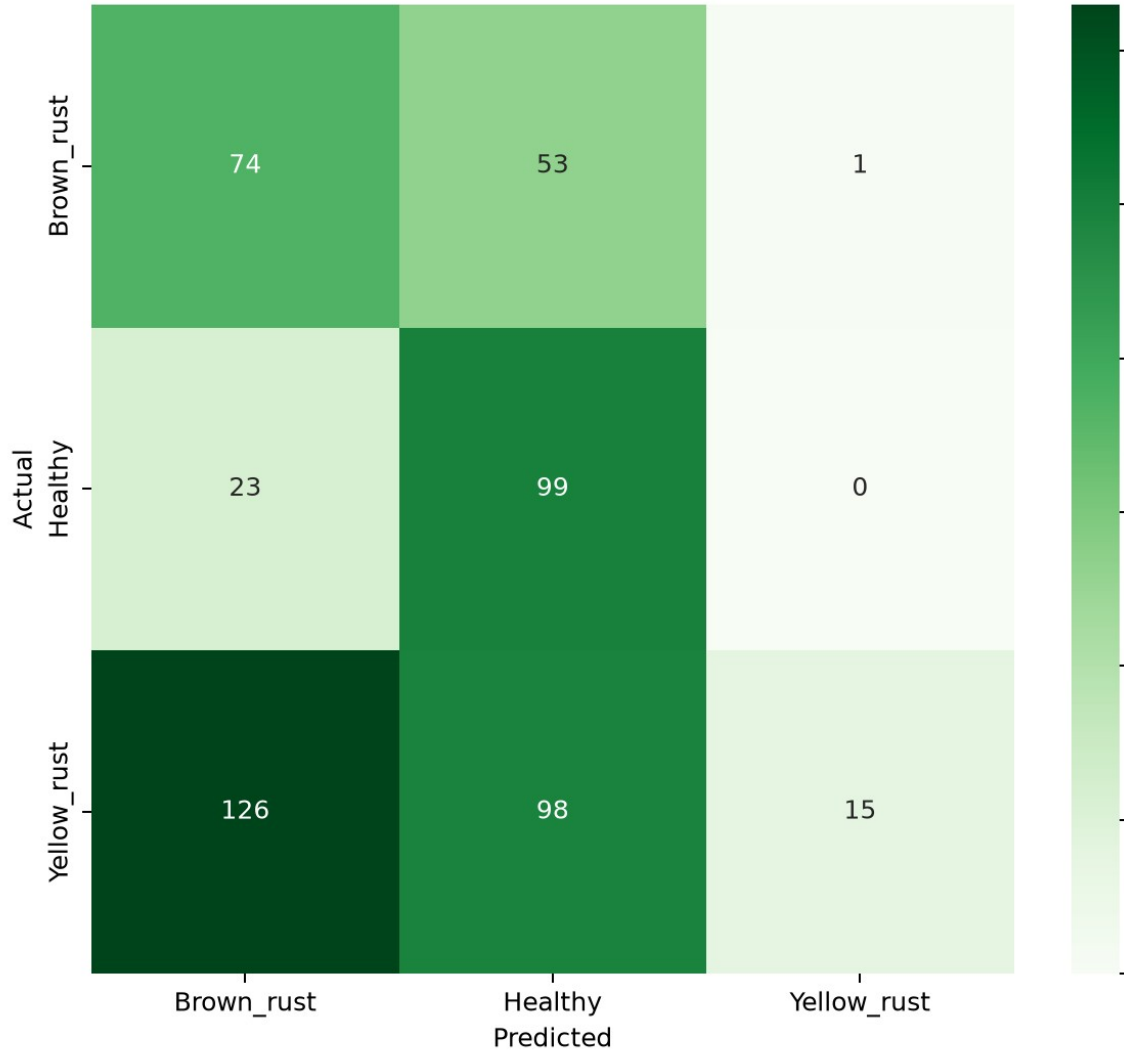


External Dataset

The accuracy did not travel

- **98.78% -** Internal validation
  - 737 images
- **38.45% -** External evaluation
  - 489 images

Only 15 of 239 yellow-rust images were classified correctly.



What changed between the datasets?

Possible differences include:

- Lighting and background.
- Camera distance and framing.
- Disease appearance.
- Class proportions and labelling practices.

This comparison shows a performance gap; it does not isolate one cause.

My next question

Does changing the dataset gives me a solution?

I compared models trained with different source mixtures.

Could including images from another source improve performance on that source—and would that improvement transfer elsewhere?

Which sources did I compare?

Original dataset

- Healthy, brown rust and yellow rust.

IARI WheatLeafRust

- Healthy and brown rust images selected for this study.

Ethiopia subset

- Healthy and yellow rust images selected for this study.

How I compared the source mixtures

Include one additional source. Exclude the other.

- **Condition A:** Include IARI; exclude Ethiopia.
- **Condition B:** Include Ethiopia; exclude IARI.

Test both models on **separate holdout images from both sources**.
Holdout images are images kept aside from training.
Repeat with **three random seeds: 42, 43 and 44**.

What did I want to find out?

- Did adding a source help on new images from **that same source**?
- Did it also help on images from the **source excluded from training**?

Example: Adding IARI

- Improvement on **IARI only** → suggests a source-specific benefit.
- Improvement on **IARI and Ethiopia** → suggests wider transfer.

Compare improvements against the model trained without the added source.

Source inclusion changed the results

Accuracy ranges across seeds 42, 43 and 44.

Performance was stronger on the included source. Transfer to the excluded source remained weak or variable.

Training Conditions	IARI Evaluation	Ethiopia Evalution
Added IARI; excluded Ethiopia	75.2–96.1%	4.9–11.5%
Added Ethiopia; excluded IARI	2.3–54.3%	85.2–90.2%

What do these results tell us?

- **Stronger performance on included sources**

  Models performed better on sources represented in training.
- **Limited transfer to excluded sources**

  Performance remained weak or varied between runs.
- **Reliable generalization is still unresolved**

  Adapting to an included source did not establish reliability elsewhere.

Limitation: The selected sources contained different class combinations, so photographic differences alone cannot explain the results.

A confident prediction can still be wrong

Confidence and Correctness are different.

Dataset label: Brown rust
Prediction: Healthy
Model confidence: 99.99%

High confidence does not guarantee a correct prediction.

What did the model focus on?



Org Image



GRAD-CAM Overlay

Dataset label: Brown rust
Model prediction: Healthy

Grad-CAM highlights regions associated with a selected class score.

A heatmap can help investigate a prediction. It cannot prove the prediction is correct.

A heatmap is a clue

A highlighted region does not establish that the model learned disease-specific features.

Stronger checks include:

- Comparing highlighted regions with expert annotations.
- Testing whether masking them changes predictions.
- Comparing those changes with random masking.

Explanation also needs evaluation.

Let’s try the research demo

- **Upload → Predict → Inspect**
- Would you trust the prediction?
- What evidence would you need?
- **Try the demo:**

devanandha-wheat-rust-ai.streamlit.app

What this project taught me

- **Ask where the score came from**
  Strong internal validation performance may not carry over to another dataset.
- **Inspect the failures**

  Overall accuracy can hide serious weaknesses in individual classes.
- **Question the explanation**

  A convincing heatmap does not prove the prediction is correct.
- **Evaluate beyond the data used for development.**

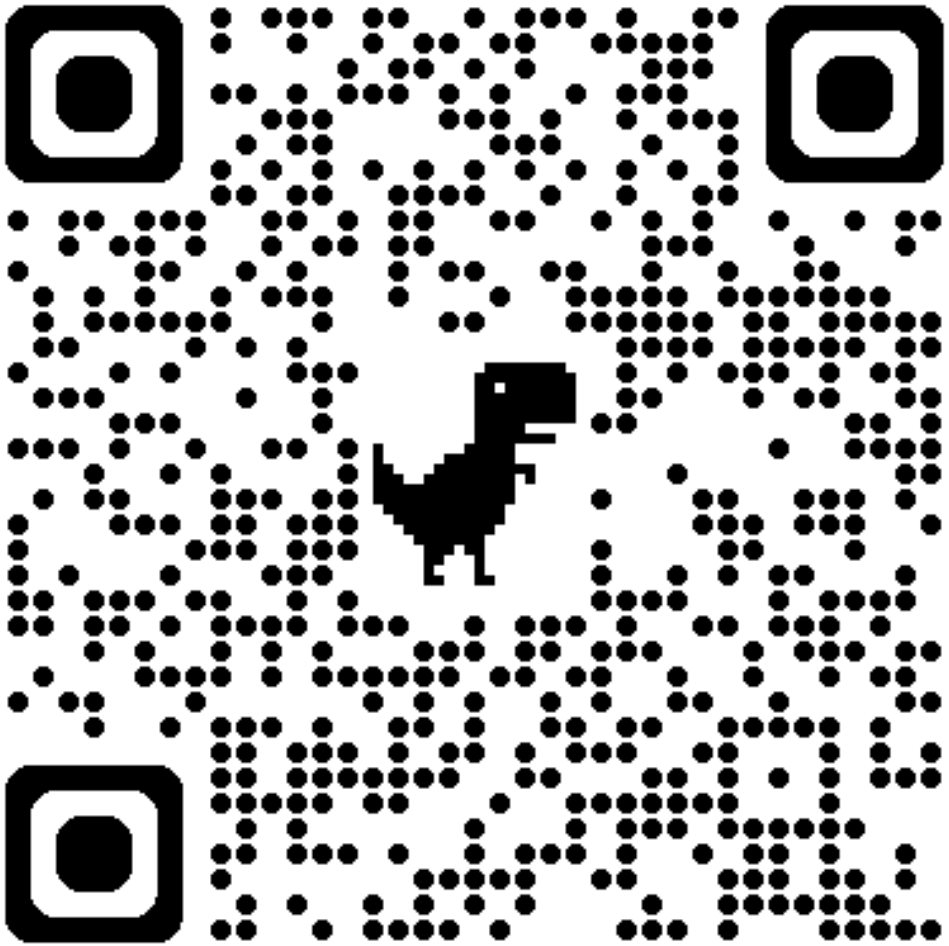
My contribution—and where I’m taking it next

- **Built**
  A wheat-rust classifier and interactive research demo**.**
- **Investigated**
  Cross-dataset performance, training-source composition and visual explanations.
- **Documented**
  Code, experimental results and failure cases for others to inspect.
- **Next research priority**

  More reliable generalization, supported by broader independent evaluation and stronger explanation checks.

Explore the Research in GitHub

- GitHub



Connect With Me on LinkedIn



Would you trust this model?

- **98.78% internal validation accuracy**
  38.45% external evaluation accuracy

What would you investigate first?

- **A** — The data and labels
- **B** — The model
- **C** — The evaluation setup

Questions, challenges and ideas welcome